Machine-Learning-Guided Performance Classification and Multiobjective Optimization of the Electrochemical Oxidation of 5-Hydroxymethylfurfural to 2,5-Furandicarboxylic Acid

Nishit Garg
[Affiliation]
[email]

Abstract

The electrochemical oxidation of 5-hydroxymethylfurfural (HMF) to 2,5-furandicarboxylic acid (FDCA) couples biomass valorization with hydrogen evolution, but the space of catalysts, electrolytes, cell architectures and operating conditions is too large for trial-and-error exploration. We compiled a literature dataset of 154 electro-oxidation records from 99 publications in 43 journals and built an end-to-end machine-learning workflow comprising leakage-controlled feature construction, K-means performance-space analysis, SMOTE/ADASYN oversampling, classifier benchmarking, SHAP interpretation and NSGA-II multiobjective optimization. Performance classes were defined directly from reported HMF conversion and FDCA yield, and every classifier was evaluated by nested cross-validation in which all records from one publication are held out together. Across 278 candidate pipelines spanning twelve model families and five resampling strategies, the selected models reach a macro-F1 of 0.610 ± 0.010 for discriminating high-performance ($\geq$90%) systems, against 0.420 for a majority-class predictor, and a macro-F1 of 0.460 (weighted-F1 0.824) on the three-class scheme of the reference methodology. Random-forest surrogates coupled with NSGA-II (population 100, 100 generations) produced a fully populated four-objective Pareto front within 0.2 min; the best-compromise candidate combines a Ni–Cu, carbon-supported, nanostructured LDH anode descriptor set with an alkaline OH^- electrolyte, TEMPO mediation and a flow-cell configuration at 9.4 °C and 1.44 V, with predicted conversion of 99.3% and FDCA yield of 96.3%. The workflow provides a reproducible, publication-aware baseline for data-driven screening of HMF electro-oxidation systems.

Keywords: 5-hydroxymethylfurfural; 2,5-furandicarboxylic acid; electrocatalysis; machine learning; imbalanced classification; nested cross-validation; NSGA-II

1 Introduction

FDCA is a bio-based diacid identified among the top value-added chemicals from biomass [1] and a direct replacement for terephthalic acid in polyethylene furanoate (PEF). Thermocatalytic HMF oxidation typically relies on noble metals, pressurized O_2 and elevated temperatures [2]. Electrochemical oxidation offers an attractive alternative: it proceeds at ambient temperature and pressure, uses water as the oxygen source, and can be paired with cathodic hydrogen evolution [3, 4]. Earth-abundant Ni, Co and Fe (oxy)hydroxides have emerged as particularly active anodes [5]. The resulting literature is large but heterogeneous: catalysts, supports, electrolytes, cell geometries and control modes (potentiostatic, galvanostatic or cell-voltage) vary from study to study.

Machine learning (ML) can extract structure from such literature datasets and guide experimental design [6]. Li et al. [7] recently demonstrated an integrated workflow for thermocatalytic HMF oxidation in which K-means clustering defines performance classes, SMOTE [8] and ADASYN [9] rebalance them, decision-tree, support-vector, random-forest and neural-network classifiers are compared, SHAP [10] explains the learned models, and a genetic algorithm searches for optimal catalyst–condition combinations.

Here we adapt this workflow to the *electrochemical* route. Three aspects of the adaptation are central. First, the electrochemical literature does not report pressure or reaction time in a comparable form, so the descriptor set and optimization objectives are re-formulated around electrode potential, temperature, electrolyte and cell architecture rather than imputed. Second, because several records usually originate

Table 1: Numerical variables of the curated dataset (154 records). n is the number of records reporting the variable.

Variable	Unit	n	Min	Median	Max
Operating conditions					
HMF concentration	mM	148	0.5	10.0	650
Electrolyte concentration	M	144	0.1	1.0	5.0
Electrode potential	V	129	0.47	1.45	2.00
Temperature	°C	91	2	25	60
Current	mA	15	0.44	50	600
Outcomes					
HMF conversion	%	117	8.0	99.9	100
FDCA yield	%	123	1.6	95.9	100
Faradaic efficiency	%	119	1.6	95.3	100
FDCA selectivity	%	91	1.6	98.0	100

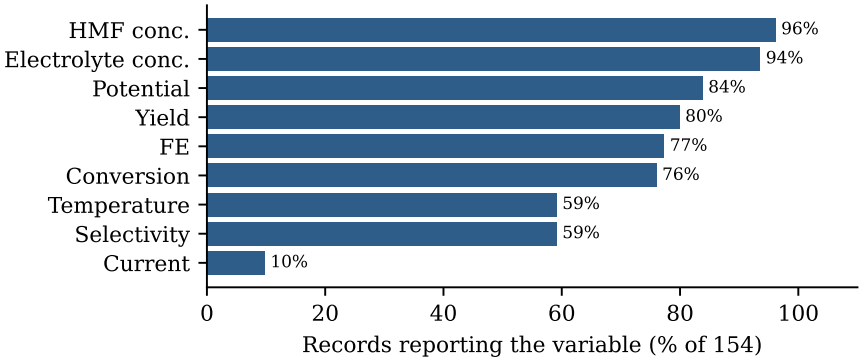


Figure 1: Fraction of the 154 records that report each variable.

from the same publication, all evaluations hold out entire publications so that reported scores measure generalization to unseen studies. Third, model selection is nested inside the cross-validation so that the reported F1 scores are not inflated by choosing the best of many pipelines on the same folds [11, 12].

2 Methods

2.1 Dataset compilation

The dataset comprises 154 records of HMF electro-oxidation extracted from 99 peer-reviewed publications in 43 journals; the most frequent sources are *Angew. Chem. Int. Ed.* (20 records), *ACS Catal.* (14) and *Appl. Catal. B* (14). Each record contains the journal, publication title, cell type, catalyst description (anode and counter electrode), electrolyte, HMF concentration, applied potential or current, temperature, TEMPO mediation, and the reported outcomes: HMF conversion, FDCA yield, faradaic efficiency (FE) and FDCA selectivity. FDCA is the target product in 148 records; the remainder report FFCA, DFF or 2,5-furandicarbaldehyde as the major product.

Table 1 summarizes the numerical variables. Outcome coverage is incomplete: FDCA yield is reported in 123 records, conversion in 117 and both in 102 (Fig. 1). Operating variables are also unevenly reported; in particular, current is available for only 15 records, reflecting the predominance of potentiostatic studies.

2.2 Feature construction

Free-text fields were converted to numerical descriptors in a single preprocessing step, implemented in `01_preprocess_data.py`.

Electrochemical regime. Potentials were extracted only from potential expressions; current-density and galvanostatic expressions were routed to a separate current variable so that applied current is never interpreted as a potential. Binary flags record whether the potential is referenced to RHE (122 records), whether the

Table 2: Performance-class schemes ($n = 123$). P is defined in Eq. (1) and Y is FDCA yield.

Scheme	Basis	Class boundaries (%)	Class sizes
C2	P	<90, ≥90	34 / 89
C3-P	P	<50, 50–80, ≥80	9 / 7 / 107
C3-Y	Y	<80, 80–95, ≥95	16 / 38 / 69
C4-Y	Y	<50, 50–80, 80–95, ≥95	9 / 7 / 38 / 69
C5-P	P	<50, 50–80, 80–90, 90–95, ≥95	9 / 7 / 18 / 22 / 67

experiment is current-controlled (16) and whether a cell voltage rather than an electrode potential is reported (13).

Catalyst. Element flags (Ni 63, Co 28, Cu 16, Fe 11, Mo 5 records, and ≤3 records each for Pd, Au, Ru, Ir, Mn, W, Ce and Ag) and material-family flags (oxide 31, carbon 35, hydroxide 17, nanoparticle 15, oxyhydroxide 12, LDH 10) were parsed from the catalyst description. Of the 74 records with a Pt flag, 71 refer to a Pt counter electrode and only 3 to a Pt-containing anode; the flag is therefore interpreted as a descriptor of the electrode assembly rather than of the anode catalyst.

Electrolyte and cell. Electrolyte concentration, an alkalinity flag (136 records), anion flags (OH^- 136, ClO_4^- 4, SO_4^{2-} 3, HCO_3^- 2, unspecified 9), TEMPO mediation (8) and cell-architecture flags (divided 90, three-electrode 22, flow 12, undivided 8) complete the descriptor set.

Engineered descriptors. For classification, these were augmented with the logarithm of HMF concentration, the HMF-to-electrolyte ratio, potential–regime interactions, noble/base-metal indicators and the number of distinct metals, giving 72 input features (`09_f1_optimization.py`).

Leakage control. Conversion, yield, FE and selectivity are never used as inputs, nor are free-text fields that describe outcomes (key feature, advantages, limitations). The publication title is used only as a grouping variable for cross-validation.

2.3 Performance-space analysis

Following the reference methodology [7], K-means clustering [13] was applied to standardized conversion and yield for the 102 complete records with $K = 3$–6 to map the structure of the performance space (`02_kmeans.py`). The clusters describe performance regions rather than catalyst families. Their visualization after SMOTE and ADASYN rebalancing is provided by `04_kmeans_oversampling.py`.

2.4 Performance classes

For each record that reports a yield, a single performance score is defined as

$$P = \begin{cases} \min(X_{\mathrm{HMF}}, Y_{\mathrm{FDCA}}) & \text{if conversion } X_{\mathrm{HMF}} \text{ is reported,} \\ Y_{\mathrm{FDCA}} & \text{otherwise,} \end{cases} \tag{1}$$

which uses the fact that yield cannot exceed conversion. This raises the number of usable records from 102 to 123. Five class schemes were evaluated (Table 2 and Fig. 2), so that conclusions do not depend on a single choice of thresholds. C3-P uses the thresholds of the reference methodology [7].

2.5 Classification pipelines

Each pipeline consists of median imputation, standardization, optional univariate feature selection (top 20 features by ANOVA F-score, or all 72), optional resampling and a classifier, implemented in scikit-learn [14] and imbalanced-learn [15]. Resampling options are none, SMOTE [8], Borderline-SMOTE [16], ADASYN [9] and random oversampling; the neighbourhood size of the synthetic samplers is reduced automatically when a minority class is smaller than the default. Twelve classifier families were considered: decision tree, random forest [17] and extra trees [18] (each with and without class weighting where applicable), gradient boosting, support-vector machine, logistic regression, k-nearest neighbours and multilayer-perceptron neural networks, each with several hyper-parameter settings. In total, 278 pipelines were evaluated per class scheme.

Resampling and feature selection are fitted inside each training fold only, so synthetic records never enter a test fold.

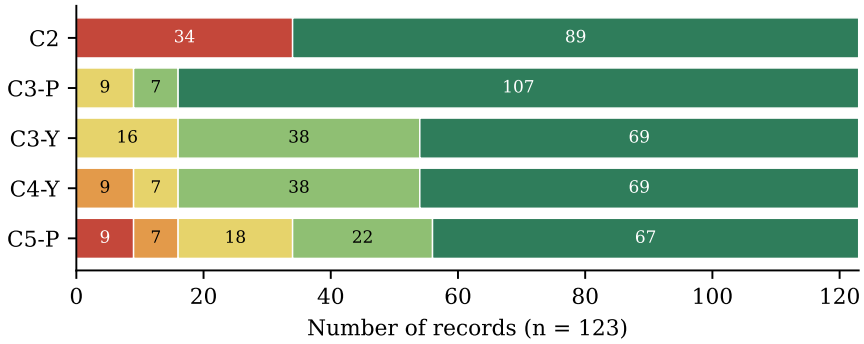


Figure 2: Class sizes for the five schemes, ordered from the lowest (left) to the highest (right) performance class.

2.6 Validation protocol

All splits are stratified by class and grouped by publication (`StratifiedGroupKFold`), so that records from a given study never appear in both training and test data.

Screening. Each pipeline is scored by five repetitions of five-fold grouped cross-validation.

Nested evaluation. Within each outer training fold, all 278 pipelines are ranked by an inner three-fold grouped cross-validation; the best pipeline is refitted on the outer training fold and scored on the held-out publications. The outer loop uses five folds and is repeated with two random seeds. Scores from this procedure are the headline results, because selecting the best of many pipelines on the same folds is optimistically biased [11, 12].

Metrics. The primary metric is macro-averaged F1, which weights all classes equally and is therefore sensitive to the minority classes; the support-weighted F1 is also reported [19]. A majority-class predictor provides the reference level for each scheme.

2.7 Surrogate models and NSGA-II optimization

For optimization, separate random-forest regressors (300 trees, minimum leaf size 2) predict HMF conversion and FDCA yield from 30 descriptors: five continuous variables (HMF concentration, electrolyte concentration, potential, temperature and current) and 25 binary flags. Binary flags with prevalence below 2% or above 98% were removed because they are unstable as free decision variables. Missing values were median-imputed. The surrogates were validated on an 80/20 split (81/21 records) and then refitted on all 102 complete records.

NSGA-II [20] was run on four objectives: maximize predicted conversion, maximize predicted yield, minimize temperature and minimize electrode potential. Temperature and potential are bounded to the observed ranges of the training records (4–60 °C and 0.47–1.90 V), and binary variables are repaired to $\{0, 1\}$ after each variation step. The population size is 100, the number of generations 100, and the crossover and mutation rates 0.90 and 0.10. Population members are evaluated in batches. Pareto-front uniformity is tracked with the spacing metric, and the standardized Euclidean distance from each candidate to its nearest training record is reported as an extrapolation diagnostic.

A single best-compromise candidate is selected from the final Pareto front by a weighted sum of min–max-normalized objectives (0.40 conversion, 0.40 yield, 0.10 temperature, 0.10 potential). This ranking is applied after optimization and is not an NSGA-II objective.

3 Results and discussion

3.1 Classification performance

Table 3 and Fig. 4 report the nested, publication-grouped scores. All five schemes exceed the majority-class baseline in macro-F1, by 0.15 (C3-P) to 0.21 (C3-Y). The binary C2 task, which separates systems reaching at least 90% performance from the rest, achieves a macro-F1 of 0.610 ± 0.010. On the three-class C3-P scheme of the reference methodology, the weighted-F1 is 0.824, reflecting accurate recognition of the dominant $\geq$80%

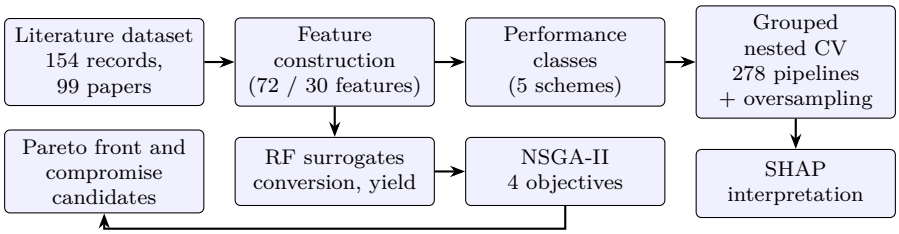


Figure 3: Overview of the workflow. The upper branch classifies reported performance; the lower branch optimizes operating conditions through random-forest surrogates and NSGA-II.

Table 3: Nested cross-validation results with publication-grouped splits (outer 5-fold, two seeds; inner 3-fold selection over 278 pipelines). Values are mean ± s.d. over seeds.

Scheme	Classes	Macro-F1	Weighted-F1	Majority-class macro-F1
C2	2	0.610 ± 0.010	0.685	0.420
C3-P	3	0.460 ± 0.058	0.824	0.310
C3-Y	3	0.445 ± 0.026	0.516	0.240
C4-Y	4	0.367 ± 0.007	0.459	0.180
C5-P	5	0.327 ± 0.002	0.426	0.141

class, while the macro-F1 of 0.460 reflects the difficulty of the two minority classes with only 9 and 7 records. Macro-F1 decreases steadily as classes become finer, from 0.610 for two classes to 0.327 for five.

Effect of model selection. The highest single-pipeline screening score on C2 is 0.763 (logistic regression with ADASYN and 20 selected features), and on C3-P it is 0.581 (extra trees with random oversampling). The corresponding nested scores are 0.153 and 0.121 lower. This gap is the optimism introduced by choosing one pipeline from 278 on the same folds, and it is why the nested scores are used throughout.

Classifier families. Table 4 lists the screening macro-F1 of the four model families used in the reference study [7], each with fixed default settings and SMOTE or ADASYN. No family is uniformly best: random forest leads on C2, C3-Y and C5-P, the neural network on C3-P and C4-Y, while the support-vector machine is weakest on every scheme with more than two classes. The pipelines most often chosen in the nested loop were similarly diverse (support-vector, logistic-regression, k-nearest-neighbour, tree-ensemble and neural-network models), which indicates that the achievable score is limited by the data rather than by the model class.

Effect of oversampling. Table 5 compares the best screening score obtained with each resampling strategy. Differences between strategies are at most 0.022 macro-F1, and the best oversampler exceeds training without resampling by at most 0.014 (C4-Y); on C2 the two are equal. Unlike in the reference study, where oversampling substantially improved generalization [7], oversampling here acts mainly as a stabilizer: the minority classes of C3-P to C5-P contain only 7–9 records, from only a handful of publications, so interpolated samples add little new information about unseen studies.

Comparison with the reference study. The reference workflow reports F1 scores of about 0.8 on 379 thermocatalytic records whose descriptors include pressure, reaction time and alkali loading, and whose classes come from K-means clusters [7]. The present dataset is a third of that size, covers a different reaction route, lacks those process variables, and is strongly skewed: 87% of records exceed 80% performance. In addition, our splits hold out whole publications, which is a stricter test than record-level splitting. The two sets of scores therefore measure different tasks and should not be compared directly.

3.2 Model interpretation

SHAP values [10] were computed for SMOTE-balanced neural-network classifiers at $K = 3$ and $K = 5$ (`06_shap_analysis.py`, `07_shap_extra_figures.py`). SHAP attributions quantify the contribution of each descriptor to the model output; they explain the fitted model and are not, by themselves, evidence of a causal chemical effect.

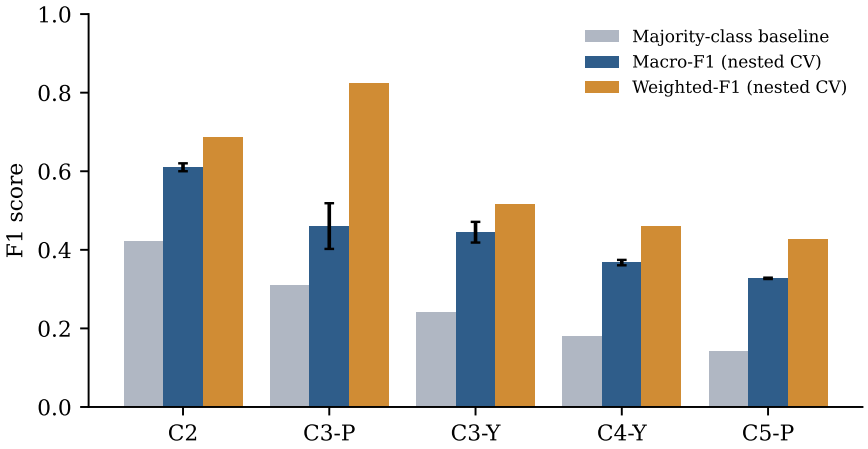


Figure 4: Nested-CV macro-F1 (error bars: s.d. over seeds) and weighted-F1 compared with a majority-class predictor for each class scheme.

Table 4: Screening macro-F1 of the reference study's four model families with fixed settings and SMOTE/ADASYN (5×5-fold grouped CV). Bold marks the best family per scheme.

Scheme	Decision tree	SVM	Random forest	Neural network
C2	0.611	0.543	**0.627**	0.597
C3-P	0.516	0.305	0.473	**0.526**
C3-Y	0.422	0.371	**0.477**	0.442
C4-Y	0.420	0.239	0.436	**0.486**
C5-P	0.282	0.236	**0.398**	0.372

3.3 Surrogate models

On the 21-record validation split, the random-forest surrogates reach an RMSE of 3.08 percentage points for conversion and 11.98 percentage points for yield (Table 6). The negative R^2 values show that the surrogates do not outperform the mean of the validation records on this split. For conversion, this follows from the very narrow target distribution (median 99.9%), in which a small absolute error is large relative to the variance. For yield, the error is substantial. The surrogates therefore define a smooth landscape for exploration, and Pareto candidates are interpreted as hypotheses for experimental testing rather than as validated optima.

3.4 Multiobjective optimization

NSGA-II completed 100 generations in 0.17 min. The front first reached the full population size of 100 at generation 7 and contained 89–100 members in every later generation. The best predicted conversion rose from 98.0% in the first generation to 99.4% and the best predicted yield from 89.8% to 97.2%; most of the gain occurred within the first 40 generations, followed by small increments up to generation 88 (Fig. 5b). The spacing metric fell from 0.150 in the first generation to 0.039–0.076 from generation 6 onward, indicating an evenly spread front (Fig. 5c). Because the problem has four objectives, the projection of the front onto the conversion–yield plane (Fig. 5a) contains points that are non-dominated only through temperature or potential.

Table 7 lists the five highest-ranked Pareto candidates. All five share an alkaline electrolyte, an RHE-referenced potential between 1.42 and 1.44 V, a Ni and/or Cu anode descriptor with nanoparticle morphology, and TEMPO mediation. Predicted conversions lie between 99.1 and 99.4% and predicted yields between 95.7 and 96.9%, at temperatures of 7.5–21.6 °C. The best-compromise candidate (score 0.863) combines Ni, Cu, LDH, carbon and nanoparticle descriptors in a flow cell with an OH^- electrolyte of 2.19 M, an HMF concentration of 238 mM and a current of 298 mA, operated at 9.4 °C and 1.44 V vs. RHE.

Three features of the front bear on experimental follow-up. First, the leading candidates lie at standardized distances of 8.2–10.0 from the nearest training record. Their HMF concentrations (185–361 mM) and currents (298–333 mA) are well above the dataset medians (10 mM and 50 mA), and current is reported in only 15 records, so these values partly reflect median-imputed regions of the descriptor space. Second, because binary descriptors are varied independently, some candidates carry no cell-architecture flag (ranks 3–5) or combine the OH^- and unspecified-anion flags (ranks 3 and 4); these are descriptor-level patterns rather than complete

Table 5: Best screening macro-F1 by resampling strategy (maximum over classifiers and feature-selection settings).

Scheme	None	SMOTE	Borderline-SMOTE	ADASYN	Random
C2	0.763	0.746	0.741	0.763	0.742
C3-P	0.579	0.581	0.575	0.578	0.581
C3-Y	0.516	0.508	0.514	0.518	0.525
C4-Y	0.484	0.491	0.477	0.498	0.489
C5-P	0.399	0.397	0.405	0.400	0.388

Table 6: Random-forest surrogate validation (80/20 split, random state 42).

Target	RMSE (pp)	R^2	Train n	Test n
HMF conversion	3.08	-0.91	81	21
FDCA yield	11.98	-0.05	81	21

experimental recipes. Third, the low predicted optimal temperatures (7.5–21.6 °C) are within the observed range of 4–60 °C for the complete records, but sit in its sparsely sampled lower part.

3.5 Limitations

The dataset contains 123 usable classification records and 102 complete regression records, drawn from studies that mix potentiostatic, galvanostatic and cell-voltage control. Temperature is missing in 41% of records and current in 90%. Minority performance classes contain fewer than ten records each, and the Pt descriptor mainly encodes the counter electrode. Explicit solvent, reaction-time and catalyst-loading descriptors are not available, so these variables cannot be optimized.

4 Conclusions

We adapted an integrated ML and genetic-algorithm workflow for HMF oxidation to the electrochemical route and evaluated it on 154 literature records from 99 publications. With performance classes defined directly from reported conversion and yield, and with publication-grouped nested cross-validation over 278 pipelines, the selected classifiers reach a macro-F1 of 0.610 for identifying high-performance systems and a weighted-F1 of 0.824 on the three-class scheme of the reference methodology, clearly above majority-class baselines. Oversampling had little effect on these scores. Random-forest surrogates coupled with four-objective NSGA-II converge rapidly and identify Ni–Cu, nanostructured, alkaline, TEMPO-mediated systems near 1.42–1.44 V vs. RHE as the most promising region of the descriptor space. Experimental testing of diverse Pareto candidates, together with chemically constrained encodings of catalyst, electrolyte and cell architecture, will allow these candidates to be confirmed and fed back into the model.

Data and code availability

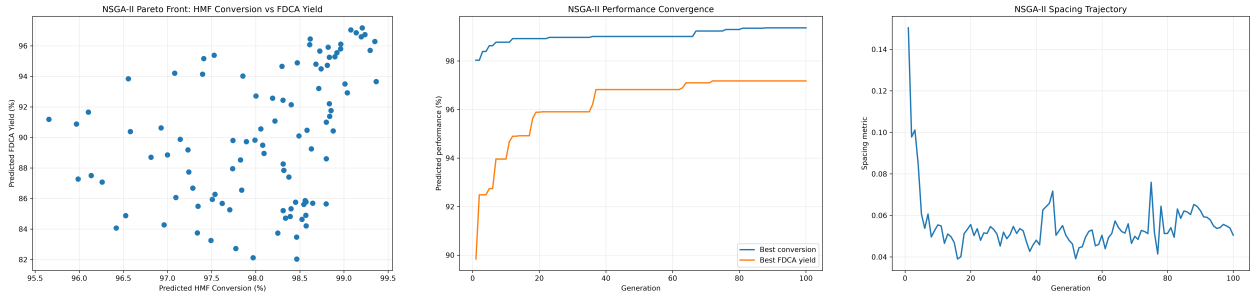
The curated dataset, preprocessing, classification and optimization scripts (`01_preprocess_data.py` to `09_f1_optimization.py`) and the result files are available in the project repository (https://github.com/Nishit-Garg-GitHub/Chemical_Ml). All results were obtained with Python, scikit-learn 1.9.1, imbalanced-learn 0.14.2, pandas 3.1.0 and NumPy 2.5.4.

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(a) **(b)** **(c)**

Figure 5: NSGA-II results. (a) Final Pareto front projected onto predicted conversion and FDCA yield. (b) Best predicted conversion and yield per generation. (c) Spacing metric per generation.

Table 7: Top five Pareto candidates ranked by the post-hoc compromise score. Conv. and Yield are surrogate predictions. $d_{\min}$ is the standardized distance to the nearest training record. All candidates use an alkaline electrolyte, RHE-referenced potential control and TEMPO.

Rank	Score	Conv. (%)	Yield (%)	T (°C)	E (V)	HMF (mM)	Electrolyte (M)	I (mA)	Metals	Material family	Cell	Anion	$d_{\min}$
1	0.863	99.35	96.29	9.39	1.441	237.9	2.19	297.8	Ni, Cu	LDH, C, NP	Flow	OH^-	9.44
2	0.846	99.30	95.71	7.51	1.441	237.9	2.19	297.8	Cu	Oxide, LDH, C, NP	Flow	OH^-	10.00
3	0.844	99.14	96.86	14.94	1.418	361.0	2.19	332.6	Ni	C, NP	—	OH^- [a]	9.08
4	0.843	99.19	96.60	14.04	1.445	223.7	2.14	306.1	Ni, Cu	NP	—	OH^- [a]	8.18
5	0.836	99.24	96.73	21.56	1.418	185.4	2.30	304.1	Ni, Cu	LDH, C, NP	—	OH^-	8.38

C = carbon, NP = nanoparticle. "—": no cell-architecture flag active. [a]The unspecified-anion flag is also active.

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